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Influence of time of tillage on soil physical attributes in potato rotations in Prince Edward Island

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Abstract

In Atlantic Canada, lack of soil cover over the cool season due to conventional primary tillage in the fall, in potato rotations, leaves the soil vulnerable to redistribution processes. The suitability of shifting primary mouldboard plough tillage operations from the fall to the spring in the potato (*Solanum tuberosum* L.) phase of a 3-year rotation (spring barley (*Hordeum vulgare* L.)–red clover (*Trifolium pratense* L.)–potato, or replacing fall mouldboard with fall chisel ploughing, was assessed over a 3-year period on Charlottetown fine sandy loam, an Orthic Podzol. A suite of soil physical attributes was used to assess the influence of primary tillage on the degree of soil loosening, permeability, and macro-structure. Potato yield and quality were also determined. Although, mouldboard ploughing provided an additional 2–10 cm of loose (i.e. penetration resistance <1.5 MPa) soil at the lower depth of the tillage zone, compared to chisel ploughing, there were no differences in soil permeability as measured by bromide movement. Soil macro-structure, or clod size, in the potato ridge just prior to harvest reflected the degree of secondary tillage rather than the influence of primary tillage. Clod size distribution in the potato ridge was not a problem for harvesting operations as clods did not exceed 20 mm in diameter. Soil permeability at the base of the potato ridge, as assessed by oxygen diffusion rate ($>55 \mu\text{g m}^{-2} \text{s}^{-1}$), air permeability, and macropore continuity (quotient of air permeability and macropore volume, $>100 \mu\text{m}^2$), and soil shear strength ($<22 \text{kPa}$) were not limiting for root growth. However soil pore space, as reflected by bulk density ($>1.25 \text{Mg m}^{-3}$) and macropore volume ($<12\%$) would pose a soil aeration limitation in this soil-type at the base of the ridge under wet soil conditions. Potato total yield and marketable yield were not influenced by differences in primary tillage over the 3-year period. Use of spring primary tillage and replacement of the mouldboard with a chisel plough, within the potato phase of a 3-year rotation, caused little change in soil physical quality compared to the conventional tillage system and appears to be a suitable conservation tillage technique for sandy loam soils in the climate of Atlantic Canada. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Conservation tillage; Soil physical quality; Humid temperate climate; Sandy loam

1. Introduction

Fall mouldboard tillage is the conventional form of primary tillage used in annual cropping systems in

eastern Canada (Cannell and Hawes, 1994; Vyn et al., 1994), especially in potato rotations in Prince Edward Island and New Brunswick. Under this system, the lack of plant cover or crop residues at the soil surface leaves the soil vulnerable to erosion processes over the cool season (Edwards and Burney, 1992). Field experimentation in Prince Edward Island has shown

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that shifting the time of tillage from the fall to the spring, and substituting chisel for mouldboard ploughing, is a useful conservation strategy for the production of spring cereals and other feed crops (Carter, 1996).

In potato production systems there is a reluctance to replace fall tillage with spring tillage. A persistent concern is that spring tillage leaves an inferior soil structural condition or macro-structure, specifically a coarse tilth in the potato ridge characterized by the presence of clods, that may adversely interfere with the sorting and selection of small tubers during harvest operations. Research elsewhere has shown that the problem of clods is mainly related to soil particle size, mineralogy, plasticity index and climate, with most of the problem confined to soils that have an excess of 10% clay (Campbell, 1980).

Generally, the chisel plough provides less soil inversion and pulverizing than the mouldboard plough (Bowen, 1981), and produces a relatively high proportion of macro-aggregates at time of tillage (Spoor and Godwin, 1984; Kouwenhoven, 1986). In addition, some studies have indicated that adoption of alternative primary tillage systems can influence soil hydraulic properties, specifically infiltration (Allmaras *et al.*, 1977). Tillage studies for the production of feed crops in Prince Edward Island have indicated that replacing mouldboard with chisel ploughing, as demonstrated for a Tignish loam, may cause soil structure and biological changes that are only periodic and transient in nature (Carter, 1996). Studies conducted elsewhere, however, have shown that in potato production the presence of both row and inter-row zones, the ridging operation, and the relatively shallow root system for potato provide the potential for both greater diversity of soil structural condition and crop response, than that found under cereals (Boone *et al.*, 1978, 1984; Ovaa and de Smet, 1984; Bouma and van Lanen, 1989).

In this study, the effect of adopting spring mouldboard ploughing and fall chisel ploughing in place of fall mouldboard ploughing, in the potato phase of a 3-year potato rotation, on specific soil physical attributes was assessed over three growing seasons in Prince Edward Island. The specific objectives were to compare the effect of differences in primary tillage on (1) the degree of soil loosening, (2) soil macro-structure in the potato ridge, (3) soil density, strength and perme-

ability immediately beneath the potato ridge, and (4) crop yield and quality.

2. Materials and methods

Over a 3-year period (1991–1993) tillage experiments were conducted at three sites at the Harrington Research Farm in central Prince Edward Island on fine sandy loams, Orthic Humo-Ferric Podzols, developed on acidic glacial till parent material (MacDougall *et al.*, 1988). Particle size distribution for the soil at the three sites is given in Table 1. The clay fraction of the A horizon is mainly composed of mica and kaolinite with lesser amounts of smectite. The majority of loam soils in Prince Edward Island, including the Charlottetown fine sandy loam, have small plasticity indices in the A horizon and the ‘field capacity’ moisture level is below the plastic limit (Carter, 1987). The soil organic C content of the A horizon ranged from 19.1 to 20.3 g kg⁻¹, and the pH ranged from 6.0 to 6.2. Prince Edward Island has a humid moisture regime and a cool to mild temperature regime as expressed by a mean annual total precipitation of 1077 mm, and a mean annual temperature of 5.9°C.

2.1. Tillage treatments

The study was part of a larger experiment initiated to assess the management of red clover in the fall prior to the potato year in a 3-year potato (i.e. barley–red clover–potato) rotation (Sanderson and MacLeod, 1998). Over a 3-year period, using a different site each year, a tillage experiment was initiated on red clover which had been undersown with barley the previous year. Each year, at the beginning (i.e. end

Table 1

Particle size distribution for the Charlottetown fine sandy loam at three experimental sites in Prince Edward Island

Site	Depth (cm)	Particle size analysis (%) ^a			
		Gravel	Sand	Silt	Clay
Site 1 (1991)	0–20	2.3	53	27	17
Site 2 (1992)	0–20	3.1	51	29	17
Site 3 (1993)	0–20	2.1	52	36	10

^aParticle size distribution: gravel (2–75 mm) sand (50–2000 µm), silt (2–50 µm), clay (<2 µm).

of red clover year and beginning of potato year) of the potato phase of the 3-year rotation, three tillage treatments were conducted using field-scale equipment: (1) fall (mid-October) mouldboard ploughing (20 cm soil depth, with 40 cm bottom plough); (2) fall (mid-October) chisel ploughing (20 cm soil depth, with spring shanks fitted with twisted shovel points spaced 30 cm apart); and spring (May) mouldboard ploughing (as in Treatment 1). Tillage was conducted at soil moisture levels below the plastic limit to prevent soil smearing or compaction. All treatments were disced (10–12 cm soil depth) twice in the spring (late May/early June) followed by harrowing prior to planting potato (cv. 'Russet Burbank'). After plant emergence, the potato rows were ridged (two to three times in growing season) using a two-row ridger, which consisted of five S-shaped tines (to loosen the soil between the rows to the 10–20 cm depth) placed ahead of a mouldboard implement that formed the ridges. The statistical design of the study was a randomized complete block experiment with four replicates ($23 \times 9 \text{ m}^2$). Nitrogen at six rates (0, 50, 100, 150, 200, and 250 kg N ha^{-1}) was banded with 65 kg P ha^{-1} and 125 kg K ha^{-1} at time of planting. All cultural practices, including pest control, were similar to those outlined in the Atlantic Canada Potato Guide (1993). Data obtained in each of the 3 years were analyzed using ANOVA techniques (Genstat 5 Committee, 1987).

2.2. Soil physical analyses

To characterize the degree of soil loosening among the three tillage systems, soil resistance to penetration was measured down the soil profile to 40 cm, at intervals of 3.5 cm, using a recording penetrometer fitted with a 13 mm diameter, 30° cone (Carter, 1988). Measurements were taken at three times: after tillage in the fall, before potato planting in the following spring, and from the centre of the potato ridge just prior to harvesting in the fall. Five to ten penetrations per plot were made at each measurement time when the soil moisture was near 'field capacity.'

To assess soil clod size distribution, large bulk soil samples ($>10 \text{ kg}$) were obtained from the potato ridges prior to harvest, at two sites per plot, and sieved (directly in the field) through a nest of sieves (25, 16, 9.5, 4.8 mm) using a Gilson Testing Screen

(ADAS, 1982). Soil moisture at time of sampling and sieving was determined by gravimetric analysis.

The permeability of the soil profile was determined using the bromide tracer technique (Carter et al., 1994). Briefly, 30 g of KBr was dissolved in 0.5 l of H_2O and applied to the surface of a 0.25 m^2 area at two sites per plot soon after tillage in the fall, for the fall mouldboard and chisel plough treatments (i.e. Treatment 1 and 2). Bromide was also applied to the soil surface of the yet unploughed Treatment 3, which served as a non-tilled control. In the spring, prior to tillage operations, soil cores (20 mm diameter) were obtained from the 0.25 m^2 areas at three depths (0–10, 10–20, and 20–30 cm). The cores at each depth were mixed, air-dried, and passed through a 2 mm sieve. To determine soil Br content, 4 g soil sub-samples were suspended in 40 ml of 0.1 M NaNO_3 , shaken for 1 h, and Br^- measured in the clear supernatant using an ion selective electrode. Bromide concentrations were determined from a standard curve.

Prior to potato harvest, five soil cores (80 mm i.d. H 80 mm) were obtained per replicate immediately under the potato ridge (adjacent to the penetrometer sampling site). The cores were wrapped in plastic in the field to prevent moisture loss, and trimmed to the edge of the cylinder in the laboratory. Soil pore volumes at a tension of 1 and 6 kPa (i.e. macropore ($50 \mu\text{m}$ equiv. diam.) volume), and air permeability, oxygen diffusion rate (ODR), and torvane shear strength on cores equilibrated at 6 kPa tension were determined sequentially according to the methods and procedures outlined in Carter (1992). Briefly, the soil core pore volumes were determined using a tension table which contained a tension media of glass beads (Topp et al., 1993). Air permeability of the soil cores was determined by using a low constant air pressure (0.25 kPa), which would prevent turbulent flow and water film disruption in the soil macropores, and a flow meter to measure the resultant steady-state air flow through the core. Oxygen diffusion rate was determined in the soil cores by use of platinum microelectrodes and an effective voltage of 0.65 mV to obtain a steady state current reading to estimate oxygen flux. Torvane shear strength was measured on the surface of each core using a shear vane (eight vanes, 19 mm wide and 29 mm long). On completion of the above measurements, the soil cores were oven dried at 105°C to determine bulk density.

2.3. Potato yield

Potato tubers were harvested, using a commercial potato harvester, from the centre two rows of each plot. Tuber yields were determined from the commercially applied rates of N (i.e. 150, 200, and 250 kg N ha⁻¹) as there were no significant N or N×tillage treatment interactions (Sanderson and MacLeod, 1998). Tuber yields are presented as total yield and marketable yield (i.e. tubers <50 mm and off-type tubers removed).

3. Results and discussion

3.1. Soil loosening profiles

Soil penetrometer resistance measured after tillage in the fall showed that mouldboard ploughing provided a looser soil profile below the 17.5 cm depth than chisel ploughing in 2 of the 3 years (Fig. 1). This is in agreement with the conclusions of Bowen (1981) and other studies on similar soil-types in Prince Edward Island (Carter, 1996). Apparent differences in soil loosening, between mouldboard and chisel ploughing, below the depth of tillage (20 cm) would be related to the elevation of the surface level by both primary tillage treatments and change in the equivalent depth of loosened soil (Campbell and O'Sullivan, 1991), as demonstrated for several soil-types under similar conditions (Carter, 1988, 1996).

Comparison of soil penetrometer resistance just prior to potato planting in the spring showed that the influence of fall tillage on the degree of soil loosening was still evident (Fig. 2). Comparison of the 1992 and 1993 fall tillage (Fig. 1) with the 1993 and 1994 spring measurements (Fig. 2) indicated that the 6 month cool period caused little overall change in the soil loosening profile. Spring mouldboard ploughing did not loosen the soil to the same degree as fall mouldboard ploughing (Fig. 2). This is probably related to wetter soil conditions in the spring, compared to the fall, at time of tillage. Comparison of soil penetrometer resistance just prior to potato harvest, taken from the top of the potato ridge, indicated that differences in soil loosening below the 24 cm depth were still evident between mouldboard and chisel ploughing, for 2 of the 3 years (Fig. 3). Very loose

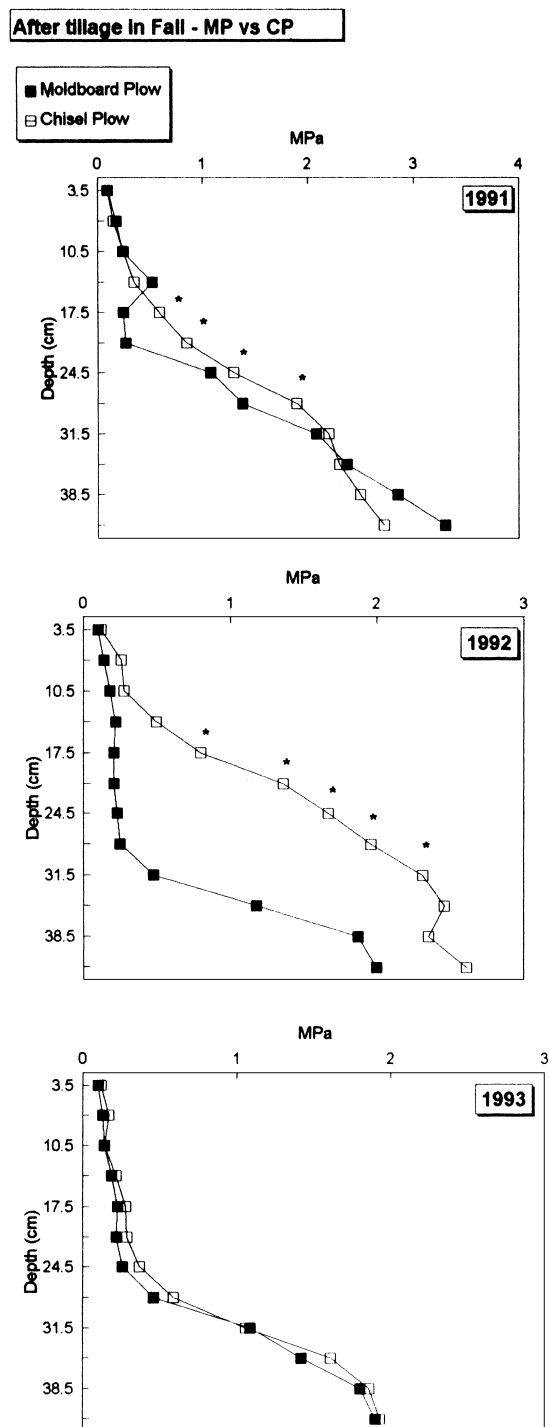


Fig. 1. Soil penetration resistance profiles after mouldboard and chisel ploughing in the fall, over a 3-year period (an asterisk indicates significant difference at $p \leq 0.05$).

Before Planting in Spring - Fall MP & CP & Spring MP

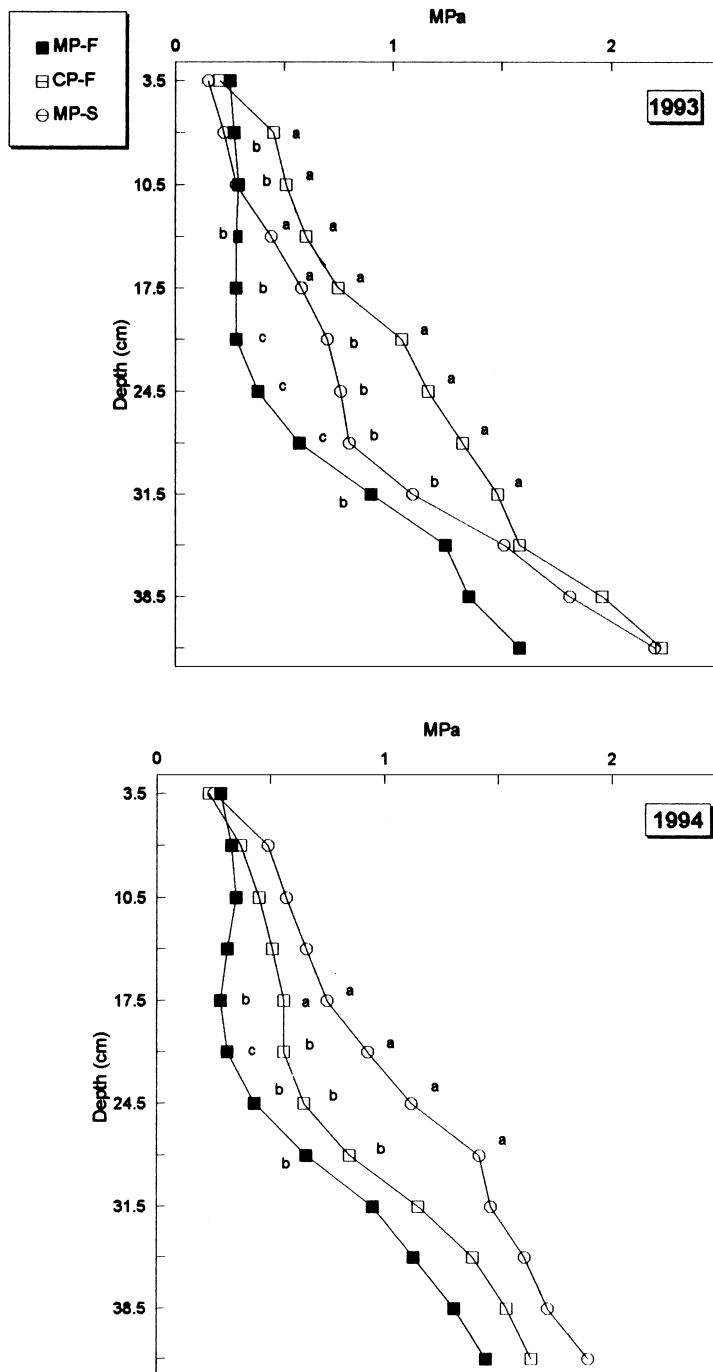


Fig. 2. Soil penetration resistance profiles just before potato planting under fall mouldboard ploughing (MP-F), fall chisel ploughing (CP-F), and spring mouldboard ploughing (MP-S), over a 2-year period. Penetration resistance values for the same depth followed by the same letter are not significantly different at $p \leq 0.05$.

prior to harvest, the depth of loose soil (≤ 1.5 MPa) ranged from 24 to 32 cm.

Potato is considered to have a slower vertical root growth and a shallower rooting depth in comparison to cereals and other crops (Ovaa and de Smet, 1984). Most of the roots tend to be confined to the topsoil. However, rooting depth is also influenced by soil-type and soil condition. Studies conducted on marine loam showed that potato roots were confined within the 0–60 cm soil depth, compared to 0–110 cm for cereals (Ovaa and de Smet, 1984). In contrast, studies on a loamy sand (Boone et al., 1978) and sandy loam (Bouma and van Lanen, 1989) found that potato rooting was confined to the 0–55 cm and 0–95 cm soil depth, respectively. Generally, a penetration resistance of 2 MPa and a macroporosity $<10\%$ are considered limiting for potato roots, in comparison to other crops (Ovaa and de Smet, 1984). Little is known about potato rooting depth and distribution in Prince Edward Island soils. However, based on the above penetration resistance criterion, rooting depth would be mainly restricted to the 0–30 cm soil depth under

chisel ploughing and the 0–40 cm depth under mould-board ploughing (Fig. 3).

3.2. Clod size distribution

Size distribution of soil clods in the potato ridge, measured prior to potato harvest, was not significantly influenced by the type of primary tillage (data not shown). This is probably due to a combination of secondary tillage and the intensive tillage involved with forming the potato ridges over the growing season. Moisture content ($\%H_2O$, w/w) of the soil in the potato ridge at time of soil sampling, which was similar among tillage treatments, was $24.6\% \pm 0.4$, $17.3\% \pm 0.4$, and $22.4\% \pm 0.5$ for 1992, 1993, and 1994, respectively. Combining the clod distribution data over treatments and years indicated that over 70% (w/w) of the soil passed through the 5 mm sieve and nearly 95% through the 16 mm sieve (Fig. 4).

A reduction in the number of soil clods from the time of ridging to harvest, due to tillage and weathering, is a common observation (Campbell, 1980). The

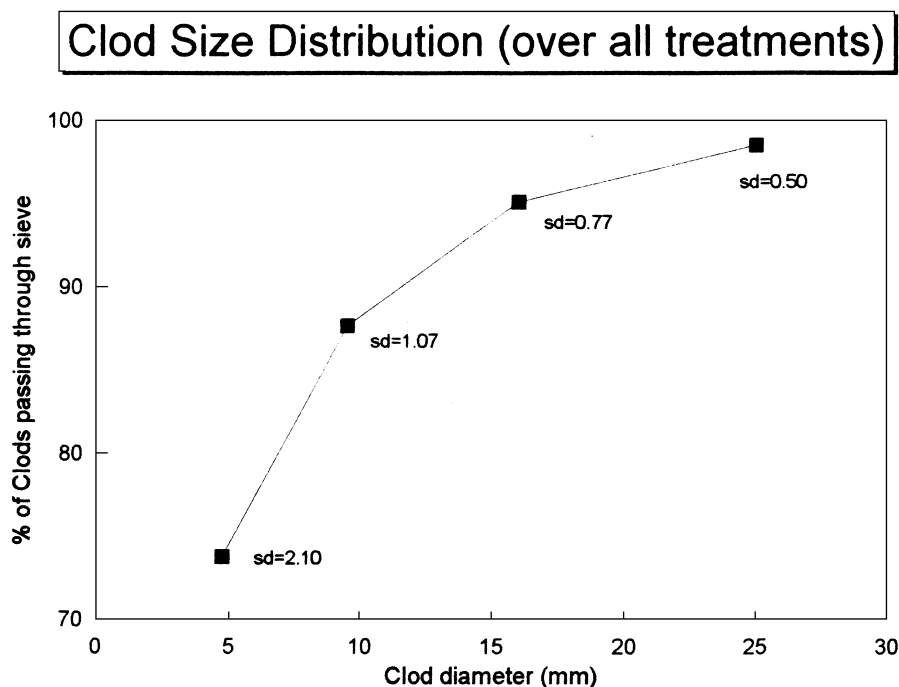


Fig. 4. Clod size distribution in the potato ridge just prior to harvest averaged over tillage treatments and the 3-year period (mean and standard deviation).

main concern, however, is the number of clods that survive passage over the rod links (20–50 mm spacing between rods) of the potato harvester. In this regard, the clod size distribution at harvest is an important soil structural indicator in potato systems. The level of clod size distribution is modified by soil-type. Kouwenhoven (1978) reported that 85% of the clods in potato ridges, in medium textured soils, were <10 mm. However, for loams with >20% clay and clay loams it is common to find 20–45 mm sized clods in the potato ridge (Campbell, 1980; Boone et al., 1984). Campbell (1980) used both the number and strength of 30–45 mm clods as an index to assess the clod problem for heavy textured soils in UK. Evidently clods at harvest did not present a problem for the present study, where nearly 100% of the soil passed through a 20 mm sieve (Fig. 4).

3.3. Soil leaching potential

Differences in primary tillage, as assessed by movement of non-reactive Br^- , can influence the depth of soil loosening (Agus and Cassel, 1992) and the degree of soil pore continuity and permeability (Bicki and Guo, 1991; Carter et al., 1994). Comparison of Br movement down the soil profile between fall mouldboard and chisel ploughing, and the undisturbed soil, over the winter period, indicated that the leaching potential was similar between tillage treatments, and between the tillage treatments and the undisturbed soil (data not shown). The combined mean bromide concentration ($\text{mg Br kg soil}^{-1}$) was relatively low (i.e. lower level of detection $\approx 1 \text{ mg Br kg soil}^{-1}$) at the 0–10 cm (mean Br: 3.0; range: 1–6) and 10–20 cm (mean Br: 3.4; range: 1–7.8) soil depth, compared to the 20–30 cm (mean Br: 16.4; range: 1.9–43.5) depth. Evidently, differences in depth of soil loosening, between primary tillage, had little influence on the soil leaching potential. These results agree with the nitrate profile studies that were also conducted on the same experiment (Sanderson and MacLeod, 1998).

3.4. Soil properties under the potato ridge

Soil physical properties under the potato ridge, measured adjacent to the penetrometer sampling site, over the 3-year period are given in Table 2. Generally, the significantly greater soil bulk density present each

year for the chisel and/or spring mouldboard plough treatments, in comparison to fall mouldboard ploughing, agrees with the penetration resistance data (Fig. 3). For the Charlottetown fine sandy loam, a soil bulk density above 1.25 Mg m^{-3} and a macropore volume (i.e. $50 \mu\text{m}$ equiv. diam.) below 12% are considered adverse for optimum soil aeration, when soil moisture is maintained near field capacity for long periods of time (Carter, 1992). The latter critical level for macroporosity is slightly higher than the commonly cited 10% (Ovaa and de Smet, 1984) due to the humid soil climate of Prince Edward Island, and potential problems with root rot disease under moist soil conditions (Carter and Johnston, 1989).

Soil shear strength was mainly within the optimum range (i.e. 18–44 kPa; Carter, 1992) for this soil-type and not considered limiting for root growth (Table 2). Again, this agrees with the penetration resistance data (Fig. 3) over the 20–25 cm soil depth (approximately at the base of the potato hill), which was below 1.5 MPa. In addition to the soil strength measurements, both ODR and air permeability (through macropores) appear non-limiting for root growth. For example, an ODR $>35 \mu\text{g m}^{-2} \text{ s}^{-1}$ is considered adequate for the root growth of many plants (Glinski and Lipiec, 1990), while yields of potato in New Brunswick were not adversely affected as long as ODR values exceeded $45 \mu\text{g m}^{-2} \text{ s}^{-1}$ in the subsoil (Sani, 1976). Deriving a macropore continuity (i.e. O) measurement by using the quotient of macropore air permeability and volume (i.e. $O = \text{air permeability} / (\text{macropore volume} \times 0.01)$) (Blackwell et al., 1990; Carter, 1992) indicated a relatively high ($O > 100 \mu\text{m}^2$) degree of soil permeability (data not shown). Values for O ranged from 232 to $788 \mu\text{m}^2$.

Overall, the soil physical characteristics at the base of the potato ridge do not appear limiting for potato root growth, especially in regard to soil strength and permeability. The soil volume data, however, suggest that under prolonged moist soil conditions the level of air-filled pore space may be limiting for adequate soil aeration.

3.5. Crop yield and quality

Potato yield was not affected by the form of primary tillage (Table 3). Tuber distribution and marketable yield, also were not influenced by tillage differences.

Table 2

Soil physical characteristics of the Charlottetown fine sandy loam under the potato ridge in three tillage treatments prior to harvesting^a

Tillage ^b	Bulk density (Mg m ⁻³)	Macropore volume (% soil volume)	Shear strength (kPa)	ODR (μg m ⁻² s ⁻¹)	Air permeability (μm ²)
<i>Site 1 (1992)</i>					
Fall-MP	1.31	7.9	19.0	67.9	50.4
Fall-CP	1.35	8.0	16.9	64.2	25.9
Spring-MP	1.30	6.8	20.1	57.8	53.6
SEM	0.021	0.79	0.99	11.54	12.10
Significance level	0.250	0.513	0.139	0.828	0.285
<i>Site 2 (1993)</i>					
Fall-MP	1.22	9.7	18.5	91.0	48.0
Fall-CP	1.33	7.6	20.6	57.7	17.6
Spring-MP	1.28	8.7	21.4	79.9	27.7
SEM	0.022	0.30	2.31	9.56	4.60
Significance level	0.036	0.008	0.678	0.116	0.009
<i>Site 3 (1994)</i>					
Fall-MP	1.20	10.6	11.8	61.0	62.5
Fall-CP	1.27	10.0	12.1	64.7	52.5
Spring-MP	1.33	9.6	19.0	91.6	47.5
SEM	0.031	0.639	1.60	12.91	10.81
Significance level	0.065	0.125	0.031	0.264	0.630

^aMacropore volume, oxygen diffusion rate (ODR), air permeability, and shear strength measured on soil cores at -6 kPa water potential (Carter, 1992).^bTillage systems: MP, mouldboard ploughing; CP, chisel ploughing.

Table 3

Potato total yield and marketable yield under three tillage systems on a Charlottetown fine sandy loam at three experimental sites in Prince Edward Island

Site	Total yield (Mg ha ⁻¹)				Marketable yield (Mg ha ⁻¹)			
	F-MP ^a	F-CP	S-MP	SEM ^b	F-MP	F-CP	S-MP	SEM
Site 1 (1992)	53.5	54.0	52.4	1.2	30.4	30.9	27.8	1.8
Site 2 (1993)	48.7	47.7	48.9	0.9	34.3	33.4	35.8	1.1
Site 3 (1994)	48.5	50.3	48.2	0.7	33.3	36.0	33.1	1.0

^aTillage system: F-MP, fall mouldboard plough; F-CP, fall chisel plough; S-MP, spring mouldboard plough.^bSEM: standard error of the mean ($n=12$, d.f.=25).

For all three tillage treatments, total yield over the 3-year period exceeded the average of 32 Mg ha⁻¹ (range of 20–45 Mg ha⁻¹) recorded for Prince Edward Island.

4. Conclusions

Shifting primary tillage operations from the fall to the spring, or replacing the mouldboard plough with a

chisel plough, had no detrimental effect on soil properties or potato yield. Thus, use of these conservation tillage techniques for potato culture, within a 3-year potato rotation, would be beneficial in Prince Edward Island, and possibly other areas in Atlantic Canada. Differences were recorded in some soil properties indicating that primary tillage can influence the soil condition, however, in most cases these differences were within the optimum range indicating that soil physical quality was not adversely affected.

Differences in depth of soil loosening and density of the soil under the potato ridge may under some circumstances (e.g. wet soil conditions) pose a problem for root growth, however, further research is needed to characterize potato root growth and distribution in these intensive production systems.

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